

Report izjednačenja prije vertikalnih pomaka

GeoTec Programm PAN, V 4.48X, 12. december 2024 14:34:33 Uhr

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12. december 2024 14:34:33 Uhr

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Selected options

option	description	value/unit
*system	Adjustment of height differences unit of measurements unit of length	meter
	unit of angle	deg
*dimen	unit for computation	mm sec
	Weight:	std,
*ankpkt	number of decimal places coordinates:	4
*ankbeo	number of decimal places observations:	4
*mce	mean centring error	0.00000
*gross_e	risc level for outlier test	0.050
*s0	a priori standard deviation	1.000
*print	printer output observations prior to adjustment residuals and data snooping obsrvation after adjustment extended panlog file	T T T T
*conf	confidence ellipses risc level	0.050
*save	saving of the adjustment results	dia

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List of initial coordinates

type of heights :

seq.no.	pt.no.	Heights [m]
1	A2 *	100.0000
2	A3 *	99.9566
3	O1 *	100.3222
4	O2 *	100.3036
5	O3 *	100.2652
6	O4 *	100.3318

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Observation prior to adjustment

group: 1: height differences shd = 1.50 [mm/km] weight: niv(std)
info:
scale:

ref.	line	inst station	target st	h-diff.	h-diff.	length	abs.
factor	standard- weight			(observed)	(computed)		term
dev.	no.			[m]	[m]	[km]	[mm]
[mm]							
	1	A2	A3	-0.0434	-0.0434	0.023	0.00
1.00	0.23 0.19E+02	A2	O1	0.3222	0.3222	0.040	0.00
1.00	0.30 0.11E+02	A2	O2	0.3033	0.3036	0.050	-0.30
1.00	0.34 0.89E+01	A3	O1	0.3660	0.3656	0.035	0.40
1.00	0.28 0.13E+02	A3	O2	0.3465	0.3470	0.040	-0.50
1.00	0.30 0.11E+02	O1	O2	-0.0186	-0.0186	0.022	-0.00
1.00	0.22 0.20E+02	O2	O3	-0.0384	-0.0384	0.008	0.00
1.00	0.13 0.56E+02	O3	O4	0.0666	0.0666	0.022	-0.00
1.00	0.22 0.20E+02	O4	O1	-0.0096	-0.0096	0.008	0.00
1.00	0.13 0.56E+02						
Height differences! Sum of abs. terms:				-0.40	[mm] sum distances[m]:	248.0	

PiD
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Residual and data snooping

group: 1: height differences shd = 1.50 [mm/km] weight: niv(std)
info:
scale:

Lfd. m.d. no. error	line EP	inst st	target st	h-diff. [m]	residual [mm]	norm. resi.	red. num.
0.77	1 0.07	A2	A3	-0.0434	-0.04	0.30	0.36
0.80	2 0.02	A2	O1	0.3222	-0.02	0.09	0.58
0.85	3 -0.06	A2	O2	0.3033	0.12	0.43	0.65
0.76	4 0.30	A3	O1	0.3660	-0.38	1.76	0.56
0.79	5 -0.25	A3	O2	0.3465	0.36	1.51	0.59
0.62	6 0.14	O1	O2	-0.0186	-0.16	0.95	0.53
0.70	7 -0.19	O2	O3	-0.0384	0.03	0.62	0.15
0.70	8 -0.13	O3	O4	0.0666	0.09	0.62	0.42
0.70	9 -0.19	O4	O1	-0.0096	0.03	0.62	0.15

Observation Seq.No. 4 has largest normalized residual : 1.76
Critical value of Tau distribution value (ALPHA = 0.050) : 1.94

Height differences! Sum residuals: 0.04 [mm] sum redundancies: 4.00000

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Variance component estimation

global test for theoretical variance :

variance of unit weight (a priori) : 1.000
variance of unit weight (a posteriori) : 1.063

test statistics : 1.063
percentile of f-distribution 2.381
degree of freedom numerator : 4
degree of freedom denominator : 999

at the chosen significance level of 0.950 the empirical variance
agrees with the theoretical variance.

a change of the a priori standard deviation is not required !

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Variance component estimation

variance component estimation :

no.	type	info	vpv	red	(s0)2	adapted sobs
1	HD		4.253	4.000	1.063 u	1.55 mm/km
Variance components per observation type:						
1	hd	Heightdifferences	4.253	4.000	1.063	

check of sum :

type	vpv	red
hd	4.2534	4.0000
total	4.2534	4.0000
check	-4.2534	4

sum lpv resp. sum degree of freedom

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Network specific indicators

Number of observations: 9

Number of directions: 0

Number of azimuths: 0

Number of zenith angles: 0

Number of distances: 0

Number of height differences: 9

Number of measured coordinates: 0

Number of coordinate differences: 0

number of network points: 6

number of fixed points: 0

number of datum points: 6

number of coordinate parameters: 6

number of additional parameters: 0

number of orientation unknowns: 0

free network parameter (rank defect) : 1

total redundancy of adjustment: 4

stand. deviation of unit weight (a priori): 1.000

stand. dev. of unit weight (a posteriori) : 1.031

number of iterations: 1

number of rank deficits: 0

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Final adjusted coordinates

Seq. No.	Point	Height [m]	dz [mm]	sz [mm]
1	A2 *	100.0001	0.07	0.15
2	A3 *	99.9566	0.03	0.14
3	O1 *	100.3223	0.05	0.10
4	O2 *	100.3035	-0.11	0.10
5	O3 *	100.2651	-0.07	0.12
6	O4 *	100.3318	0.02	0.12

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Point related measures of precision

confidence intervall : alpha = 0.050 percentile of t-distribution : 2.13				
seq. no.	point	u [m]	o [m]	
1	A2 *	99.99976	100.00039	
2	A3 *	99.95633	99.95693	
3	O1 *	100.32204	100.32246	
4	O2 *	100.30328	100.30371	
5	O3 *	100.26487	100.26539	
6	O4 *	100.33156	100.33207	

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Observations after adjustment

group: 1: height differences shd = 1.50 [mm/km] weight: niv(std)								
info:								
scale:								
ref. residual no.	line	inst st	target st	h-diff. (adjusted) [m]	stand. abw. [mm]	h-diff. (computed) [m]	[m]	
14	1	A2	A3	-0.0434	0.19	-0.0434	-0.89E-	
	2	A2	O1	0.3222	0.20	0.3222	-0.89E-	
15	3	A2	O2	0.3034	0.21	0.3034	-0.39E-	

14							
	4	A3	O1	0.3656	0.19	0.3656	0.80E-
14	5	A3	O2	0.3469	0.20	0.3469	0.49E-
14	6	O1	O2	-0.0188	0.16	-0.0188	-0.30E-
14	7	O2	O3	-0.0384	0.13	-0.0384	0.45E-
14	8	O3	O4	0.0667	0.17	0.0667	-0.20E-
14	9	O4	O1	-0.0096	0.13	-0.0096	0.55E-
15	Maximum linearization error: 0.885403E-14 Program PAN(INTEL) finished. Elapsed cpu time: 0.03 sec!						

Report izjednačenja nakon vertikalnih pomaka

GeoTec Programm PAN, V 4.48X, 12. december 2024 14:34:52 Uhr

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Selected options

option	description	value/unit
*system	Adjustment of height differences	
	unit of measurements	
	unit of length	meter
*dimen	unit of angle	deg
	unit for computation	mm sec
	Weight:	std,
*ankpkt	number of decimal places coordinates:	4
*ankbeo	number of decimal places observations:	4
*mce	mean centring error	0.00000
*gross_e	risc level for outlier test	0.050
*s0	a priori standard deviation	1.000
*print	printer output	
	observations prior to adjustment	T
	residuals and data snooping	T
	observation after adjustment	T
*conf	extended panlog file	T
	confidence ellipses	
	risc level	0.050
*save	saving of the adjustment results	dia

PiD
dd.mm.yyyy

List of initial coordinates

type of heights :

seq.no.	pt.no.	Heights [m]
1	A2 *	100.0000
2	A3 *	99.9566
3	O1 *	100.3222
4	O2 *	100.3036
5	O3 *	100.2652
6	O4 *	100.3318

PiD
dd.mm.yyyy

Observation prior to adjustment

group: 1: height differences shd = 1.50 [mm/km] weight: niv(std)
info:
scale:

ref.	line	inst station	target st	h-diff.	h-diff.	length	abs.
factor	standard- weight			(observed)	(computed)		term
dev.	no.			[m]	[m]	[km]	[mm]
[mm]							
	1	A2	A3	-0.0433	-0.0434	0.023	0.10
1.00	0.23 0.19E+02	A2	O1	0.3228	0.3222	0.040	0.60
1.00	0.30 0.11E+02	A2	O2	0.3034	0.3036	0.050	-0.20
1.00	0.34 0.89E+01	A3	O1	0.3661	0.3656	0.035	0.50
1.00	0.28 0.13E+02	A3	O2	0.3462	0.3470	0.040	-0.80
1.00	0.30 0.11E+02	O1	O2	-0.0193	-0.0186	0.022	-0.70
1.00	0.22 0.20E+02	O2	O3	-0.0397	-0.0384	0.008	-1.30
1.00	0.13 0.56E+02	O3	O4	0.0659	0.0666	0.022	-0.70
1.00	0.22 0.20E+02	O4	O1	-0.0075	-0.0096	0.008	2.10
1.00	0.13 0.56E+02						
Height differences! Sum of abs. terms:				-0.40	[mm] sum distances[m]:	248.0	

PiD
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Residual and data snooping

group: 1: height differences shd = 1.50 [mm/km] weight: niv(std)
info:
scale:

Lfd. m.d. no. error	line EP	inst st	target st	h-diff. [m]	residual [mm]	norm. resi.	red. num.
0.77	1 -0.15	A2	A3	-0.0433	0.09	0.49	0.36
0.80	2 0.11	A2	O1	0.3228	-0.16	0.54	0.58
0.85	3 -0.01	A2	O2	0.3034	0.01	0.03	0.65
0.76	4 0.19	A3	O1	0.3661	-0.24	0.90	0.56
0.79	5 -0.29	A3	O2	0.3462	0.43	1.44	0.59
0.62	6 -0.06	O1	O2	-0.0193	0.07	0.33	0.53
0.70	7 -0.62	O2	O3	-0.0397	0.11	1.66	0.15
0.70	8 -0.42	O3	O4	0.0659	0.31	1.66	0.42
0.70	9 -0.62	O4	O1	-0.0075	0.11	1.66	0.15

Observation Seq.No. 9 has largest normalized residual : 1.66
Critical value of Tau distribution value (ALPHA = 0.050) : 1.94

Height differences! Sum residuals: 0.72 [mm] sum redundancies: 4.00000

PiD
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Variance component estimation

global test for theoretical variance :

variance of unit weight (a priori) : 1.000
variance of unit weight (a posteriori) : 1.645

test statistics : 1.645
percentile of f-distribution 2.381
degree of freedom numerator : 4
degree of freedom denominator : 999

at the chosen significance level of 0.950 the empirical variance
agrees with the theoretical variance.

a change of the a priori standard deviation is not required !

PiD
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Variance component estimation

variance component estimation :

no.	type	info	vpv	red	(s0)2	adapted sobs
1	HD		6.578	4.000	1.645 u	1.92 mm/km
Variance components per observation type:						
1	hd	Heightdifferences	6.578	4.000	1.645	

check of sum :

type	vpv	red
hd	6.5783	4.0000
total	6.5783	4.0000
check	-6.5783	4

sum lpv resp. sum degree of freedom

PiD
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Network specific indicators

Number of observations:	9
Number of directions:	0
Number of azimuths:	0
Number of zenith angles:	0
Number of distances:	0
Number of height differences:	9
Number of measured coordinates:	0
Number of coordinate differences:	0
number of network points:	6
number of fixed points:	0
number of datum points:	6
number of coordinate parameters:	6
number of additional parameters:	0
number of orientation unknowns:	0
free network parameter (rank defect):	1
total redundancy of adjustment:	4
stand. deviation of unit weight (a priori):	1.000
stand. dev. of unit weight (a posteriori) :	1.282
number of iterations:	1
number of rank deficits:	0

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Final adjusted coordinates

Seq. No.	Point	Height [m]	dz [mm]	sz [mm]
1	A2 *	100.0005	0.45	0.18
2	A3 *	99.9572	0.64	0.17
3	O1 *	100.3231	0.89	0.12
4	O2 *	100.3039	0.26	0.13
5	O3 *	100.2643	-0.93	0.15
6	O4 *	100.3305	-1.32	0.15

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Point related measures of precision

confidence intervall : alpha = 0.050 percentile of t-distribution : 2.13				
seq. no.	point	u [m]	o [m]	
1	A2 *	100.00006	100.00084	
2	A3 *	99.95686	99.95761	
3	O1 *	100.32283	100.32335	
4	O2 *	100.30359	100.30413	
5	O3 *	100.26395	100.26460	
6	O4 *	100.33016	100.33080	

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Observations after adjustment

group: 1: height differences shd = 1.50 [mm/km] weight: niv(std)							
info:							
scale:							
ref. residual no.	line	inst st	target st	h-diff. (adjusted) [m]	stand. abw. [mm]	h-diff. (computed) [m]	[m]
14	1	A2	A3	-0.0432	0.23	-0.0432	-0.46E-
14	2	A2	O1	0.3226	0.25	0.3226	0.42E-
14	3	A2	O2	0.3034	0.26	0.3034	0.16E-
14	4	A3	O1	0.3659	0.24	0.3659	0.88E-

14	5	A3	O2	0.3466	0.25	0.3466	0.62E-
14	6	O1	O2	-0.0192	0.20	-0.0192	-0.26E-
14	7	O2	O3	-0.0396	0.16	-0.0396	0.39E-
15	8	O3	O4	0.0662	0.22	0.0662	-0.92E-
14	9	O4	O1	-0.0074	0.16	-0.0074	0.11E-
13	Maximum linearization error: 0.114821E-13 Program PAN(INTEL) finished. Elapsed cpu time: 0.02 sec!						

Report definitivne analize

GeoTec Programm DEFANA V2.10

12. december 2024 14:37:45 Uhr

```

#####
###   ##   #####
###   ##   ###   ###   ###   ##   #####
###   ##   ###   ###   ###   ##   ##   ###   ##
###   ##   #####   #####   ###   ##   ##   ###   ##
###   ##   ###   ###   #####   ###   ##   ##   #####
###   ##   ###   ###   ###   ##   ###   ###   ##
#####   #####   ##   ###   ##   ##   ##   ##

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Output of the options

Information about both epochs

```

Identical points      :      6
Reference points      :      6
Degrees of freedom    :      8
Standard deviation    :      1.16

```

```

Datum defect
Z-translation        : T

```

Characteristics of epoch 1

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Points                :      6
Degees of freedom    :      4
Standard deviation    :      1.03
File                  : inp\vert_pomaci_prije.gll

```

Characteristics of epoch 2

```

PiD
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Points                :      6
Degees of freedom    :      4
Standard deviation    :      1.28
File                  : inp\vert_pomaci_poslije.gll

```

Output of selected options

```

Unit of computation   : MM
Risc level             : 0.050
Two step analysis     : T
Result file           : T
Plot file              : T
Extended output       : T

```

Faktor S0	:	1.00
Global individual movement	:	.0000

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Indential points

Seq. No.	Point	Appr Hght Z [m]	Epoch 1 dZ [MM]	Epoch 2 dZ [MM]	Indivi move [MM]
1	A2	100.0000	0.075	0.451	0.000
2	A3	99.9566	0.033	0.637	0.000
3	O1	100.3222	0.053	0.894	0.000
4	O2	100.3036	-0.106	0.262	0.000
5	O3	100.2652	-0.073	-0.926	0.000
6	O4	100.3318	0.019	-1.318	0.000

PiD
PiD-----
Analysis within reference points (backward)

Global test:	Run No:	1
Size of Qxx - matrix:	Submatrices	
Standard deviation of unit weight:		1.16
Datum points:		6
Degrees of freedom nom. / denom.:	5 /	8
Percentile of F-dist. / critical value:	3.69 <	26.15

Result of the analysis:

H0 is rejected : significant movements in the network !
The analysis within the reference points is being continued !

Result of the localisation : Discrepancies of the reference points

Seq.No.	Point	dz [MM]	Dis [MM]	T	Result
1	A2	0.38	0.38	0.66	
2	A3	0.60	0.60	1.84	
3	O1	0.84	0.84	6.24	
4	O2	0.37	0.37	1.14	
5	O3	-0.85	0.85	4.63	
6	O4 *	-1.34	1.34	11.64	Pt moved

PiD
PiD-----
Analysis within reference points (backward)

Global test: Run No: 2
Size of Qxx - matrix: Submatrices
Standard deviation of unit weight: 1.16
Datum points: 5
Degrees of freedom nom. / denom.: 4 / 8
Percentile of F-dist. / critical value: 3.84 < 14.97

Result of the analysis:

H0 is rejected : significant movements in the network !
The analysis within the reference points is being continued !

Result of the localisation : Discrepancies of the reference points

Seq.No.	Point	dz [MM]	Dis [MM]	T	Result
1	A2	0.11	0.11	0.07	
2	A3	0.34	0.34	0.75	
3	O1	0.57	0.57	3.67	
4	O2	0.10	0.10	0.11	
5	O3 *	-1.12	1.12	10.37	Pt moved

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PiD

Analysis within reference points (backward)

Global test: Run No: 3
Size of Qxx - matrix: Submatrices
Standard deviation of unit weight: 1.16
Datum points: 4
Degrees of freedom nom. / denom.: 3 / 8
Percentile of F-dist. / critical value: 4.07 > 2.07

Result of the analysis:

H0 is accepted : no significant movements in the network!
End of the analysis within the reference points !

Result of the localisation : Discrepancies of the reference points

Seq.No.	Point	dz [MM]	Dis [MM]	T	Result
1	A2	-0.17	0.17	0.26	
2	A3	0.06	0.06	0.03	
3	O1 *	0.29	0.29	1.31	
4	O2	-0.18	0.18	0.47	

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Analysis within object points (forward)

Localisation: 1. step
Size of Qxx matrix: full
Standard deviation of unit weight: 1.16
Object points: 2

Result of the localisation : discrepancies of the object point

Seq.No.	Point	dz [MM]	Dis [MM]	SDis [MM]	Result
1	O3 *	-1.40	1.40	0.22	Pt moved
2	O4	-1.88	1.88	0.22	

Global test:	Run No:	1
Size of Qxx - matrix:	Submatrices	
Standard deviation of unit weight:		1.16
Datum points:		5
Degrees of freedom nom. / denom.:	4 /	8
Percentile of F-dist. / Critical value:	3.84 <	14.97

Result of the analysis:

H0 is rejected : significant movement in network !
End of the deformation analysis for the object points

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PiD

Analysis within object points (F-test)

Localisation :	Single point test
Size of Qxx matrix :	full
Standard deviation of unit weight :	1.16
Datum points :	4
Degrees of freedom nom. / denom. :	1 / 8
Percentile of F-distribution :	5.33

Result of the localisation : discrepancies of the object points

Seq.No.	Point	dz [MM]	Discr [MM]	T	Result
1	O3	-1.40	1.40	41.46	Pt moved
2	O4	-1.88	1.88	76.84	Pt moved

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Result: discrepancies

Type of point :	Reference points
Size of Qxx matrix :	Submatrices

Seq.No.	Point	dz [MM]	Dis [MM]
1	A2	-0.17	-0.17
2	A3	0.06	0.06
3	O1	0.29	0.29
4	O2	-0.18	-0.18